



Repeating a dose of sucrose for heel prick procedure in preterms is not effective in reducing pain: a randomised controlled trial

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Received: 26 June 2019 / Revised: 12 October 2019 / Accepted: 15 October 2019 / Published online: 15 November 2019
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Abstract

Oral sucrose is included in almost all recommendations for treatment of pain in newborns, but evidence if multiple doses might be more effective than a single standard dose is lacking. We designed a single-centre, double-blind, randomised, controlled trial. We enrolled preterm infants needing the heel prick procedure. Each enrolled infant was randomised to receive a single standard dose of sucrose 2 min before or a double dose of sucrose 2 min before, and 30 s after heel prick. Primary outcome was the efficacy of the two interventions tested by the premature infant pain profile-PIPP scale obtained at 30 s, 60 s, and 120 s after heel prick. Secondary outcome was the evaluation of the concordance between the PIPP scale and other pain scores more feasible in clinical practice. Seventy-two infants were randomised. No difference in pain perception as measured by the PIPP scale was found between the groups: median PIPP values 4.0(IQR 3.0–4.0) vs 3.0(IQR 3.0–4.0) at baseline; 6.0(IQR 5.0–10.0) vs 6.0(IQR 4.0–8.5) at 30 s; 6.0(IQR 4.0–7.0) vs 5.0(IQR 4.0–8.5) at 60 s and 5.0(IQR 4.0–7.0) vs 5.0(IQR 4.0–7.5) at 2 min, in the experimental and standard treatment groups, respectively ($p = 0.9020$). There was no correlation between PIPP scores and other pain scales. **Conclusion:** We do not recommend doubling the dose during heel prick.

What is Known:

- Oral sucrose is included in almost all international position papers and recommendations for the treatment of mild to moderate pain in newborns, associated with non-nutritive sucking and facilitated tucking
- Premature infant pain profile (PIPP) scale is the gold standard for evaluation of pain in preterms but it is difficult to use in clinical practice

What is New:

- Repeating a dose of 24% sucrose is not effective in reducing pain during the recovery phase of a skin breaking procedure
- Other pain scales, easier to use in clinical practice, are not comparable with PIPP for the evaluation of procedural pain in preterms

Keyword Sucrose · Pain · Heel prick · Preterm infant · Non-nutritive sucking · Facilitated tucking

Communicated by Patrick Van Reempts

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Abbreviations

HP	Heel prick
GA	Gestational age
NICU	Neonatal intensive care unit
PIPP	Premature infant pain profile
FLACC	Faces, legs, activity, cry and consolability
VAS	Visual Analog Scale
NIPS	Neonatal Infant Pain Scale
SD	Standard deviation
NNS	non-nutritive sucking
FT	Facilitated tucking

Introduction

During the prolonged hospitalisation in the Neonatal Intensive Care Units (NICUs), preterm babies experience repeated, painful procedures.

Although in the last few years, awareness of premature feelings of pain has allowed an improvement in neonatal pain management, the frequency of painful procedures performed with insufficient analgesia remains high [7]. It has been demonstrated that a multidimensional approach, comprised of pharmacological (e.g. analgesic drugs) and non-pharmacological interventions (e.g. oral sucrose, breastfeeding, non-nutritive sucking [NNS], facilitated tucking [FT], kangaroo care, and swaddling) is effective at reducing pain [1, 5, 6, 10, 16, 21].

The administration of oral sucrose has been extensively used as an analgesic during minor procedures in neonates, and it is now included in almost all international position papers and recommendations for treatment of mild to moderate procedural pain in newborns [1, 5, 6, 8, 14, 21].

There is high-quality evidence for the beneficial effect of the association of 24% sucrose with NNS to reduce pain in premature during heel prick (HP) and moderate quality of evidence that sucrose in combination with other non-pharmacological interventions, such as FT, is more effective than sucrose alone [1, 7].

A wide range of dosages of sucrose 24% (0.1–2 mL) have been tested in different trials (most commonly 0.3 mL for preterm and 0.5 mL for term infants), but the optimal dose has not been identified yet [21].

It is still also unknown if multiple doses before and during the procedure might be more effective than the single standard dose before HP. The evidence of a positive association between the number of painful procedures and an increased risk of poor neurodevelopmental outcomes in preterm infants is well documented [10], but concerns are now being raised over the risks associated with the use of multiple doses of sucrose, as the long-term effects of repeated doses are still unknown [12].

The evaluation of pain during painful procedures relies on a dedicated pain scale (the premature infant pain profile [PIPP] scale), which should be performed at 30, 60, and 120 s after the HP [20]. The use of the PIPP scale in clinical practice might be difficult: too many parameters must be evaluated for 30 s after the painful stimulus by a single operator (changes in heart rate, saturations, facial actions). Even the revised version of the PIPP scale has the same limitation [9].

Other pain scales that are easier to use in clinical practice are available to evaluate pain in newborns [15, 17], but no other pain scales have been compared to PIPP for the evaluation of pain in preterm infants.

Our primary aim was to test whether a repeated dose of sucrose is more effective than a standard dose alone in control HP pain evaluated with a standardized pain scale (PIPP); we also wanted to understand if a previous standardized dose of oral sucrose was adequate for pain control in premature infants and to evaluate the concordance and reliability of different pain scales relative to the PIPP scale.

Methods

Trial design, patients, and setting

This study was a randomised, double-blind, controlled trial (RCT) conducted at the NICU of the Department of Woman and Child Health, University Hospital of Padua, between August 2016 and October 2017.

All preterm infants admitted to the NICU (23–36 + 6 weeks of gestation at birth) with a postnatal age less than 28 days or before reaching 40 weeks corrected postmenstrual age undergoing HP for a routine test, were eligible for enrolment.

We excluded infants with significant congenital or clinical abnormalities, as well as those suffering from birth asphyxia (Apgar score less than 5 at 5 min and/or arterial cord pH less than 7.00), myopathies or a neurological disorder (severe IVH or periventricular leukomalacia) interfering with pain expression and/or born from a mother addicted to substance abuse or in treatment for a neurological disorder. Arterial line, analgesic and sedative medications and acute illness (necrotizing enterocolitis, sepsis), present at the enrolment were also exclusion criteria, as well as current participation in any other clinical study of an investigational medicinal product.

The study was approved by the Hospital Ethics Committee (reference number 3714/AO/2016 NEOSUCROSE) and registered in [Clinicaltrials.gov](https://clinicaltrials.gov) (NCT 02859376) before the first patient was enrolled.

Interventions

The participants received a first dose of sucrose 24% 2 min before the procedures and a second dose of sucrose or placebo

during the HP procedure. In the experimental treatment group (A), both syringes 1 and 2 contained sucrose (24%, 0.3 mL, 0.07 g of sucrose each) for infant ≤ 1000 g or 0.5 mL for infants > 1000 g (0.12 g of sucrose each); meanwhile, in the standard treatment group (B), infants were given a dose of sucrose (24%, 0.3 mL or 0.5 mL) in syringe 1 and the same quantity of sterile water in syringe 2.

The HP was performed by the infant's nurse according to a standardised unit check-list following three phases: (1) Baseline phase, (2) heel prick phase, and (3) recovery phase. Data collection started during the baseline phase, before any manipulation. Sucrose was given after the basal algometric evaluation, directly on the tongue with the syringe, 2 min before the procedure together with a pacifier for NNS and FT, which continued to be performed during all of the procedure by a nurse trained with a standardised procedure who assisted the HP. After accurate skin preparation, the HP was performed using a blade-shaped automatic lancet with "arched" cutting edges of two different measures for an infant $\leq$ (cutting width of 1.40 mm, 0.65 deep) or $>$ than 1000 g (cutting width of 1.75 mm, 0.85 mm deep). Thirty seconds after the HP, the second syringe was administered, containing a dose of sterile water or sucrose depending on the treatment group. Meanwhile, a blood sample for capillary blood gas analysis was withdrawn, and haemostasis was done. In the third recovery phase, lasting 2 min after HP, the neonate was followed until he/she returned to the baseline status. Using a standardised procedure, all phases of the procedure were videotaped.

Pain response, the dependent variable, was measured using the PIPP scale [20]. The pain evaluations were performed at the baseline time, 30 s, 60 s, and 120 s after the HP. Greater changes in the physiologic and behavioural items indicated increased pain, whereas a score greater than 12 was considered moderate-severe pain. The HPs were also rated using other pain scales more feasible in the clinical practice and indicate in Italy for their use in newborn (FLACC: Faces, Legs, Activity, Cry and Consolability; indirect VAS 10 cm Visual Analogic Scale; NIPS: Neonatal Infant Pain Scale), to compare the performance of these pain scales with the goal standard PIPP [9, 15, 17, 18, 20]. As each pain scale has a different maximum attributable score, which did not make a direct comparison between scores possible, only assuming classes of pain grading (i.e. no pain, mild, moderate, severe for each scale) make them comparable [20]. After categorizing each scale for the range score corresponding to an absent-mild pain and a moderate-severe pain according to the literature, the two researchers (TM and AA) responsible for the blind evaluation of the pain, performed an independent assessment of the videos after having attended a standardized instruction session performed by an expert evaluator on the modality to perform the rating, which was a neonatologist expert in sedoanalgesia and evaluation of pain scales.

Recruitment and randomisation

Written informed consent was obtained from the infant's caregiver prior to enrolment. The treatment allocation schedule was performed using a computer-generated randomisation list. Group assignment was written on cards and placed in sequentially numbered, sealed opaque envelopes by an independent assistant. Immediately before the procedure, a research nurse, opened the next envelope in the sequence and prepared the two syringes numbered 1 and 2. The syringes used to administer sucrose were also shielded from view by the research nurse from the bedside nurse and video recording. No other study personnel had access to the treatment allocation.

Primary outcome

The primary outcome of our study was to investigate if a second dose of sucrose (24%) administered during HP, associated with NNS and FT was more effective in pain control than a single standard dose of sucrose (24%).

Secondary outcomes

Secondary outcomes included: the inter-rater reliability between raters; the efficacy of the standardised dosages; possible adverse events related to sucrose administration; and the concordance with PIPP of different pain scales (NIPS, FLACC, indirect VAS) easier to use in clinical practice.

Blinding

Investigators, clinicians, and caregivers performing the procedures were masked to the infant group assignment.

Sample size, power, and statistical methods

The sample size estimate was based on the primary endpoint: the efficacy of experimental treatment evaluated with the PIPP scale. While a preliminary calculation fixed the sample size in 144 patients (72 per group), a careful review of the statistical methodology with our University Statistician changed the number needed of patients included as follow. A sample size of 31 patients per arm allows us to show a clinically significant difference of at least three points on the means of the PIPP scale having a standard deviation of four and considering a two-sided 5% significance level and a power of 80% (Mann-Whitney test, PASS11, NCSS, LLC, Kaysville, Utah, USA). The sample size was increased to 36 per group considering a 5% of dropouts.

Concordance between raters in the evaluation of pain scales was measured by the concordance correlation coefficient (CCC) for repeated measures accompanied by the 95%

confidence interval (95% CI) calculated using the U statistic method [4].

The patient's demographic and clinical characteristics and clinical outcomes have been described in the treatment groups by mean and standard deviation or median, minimum and maximum for quantitative variables, number and percentage of patients in each category for qualitative variables.

The patient's demographic and pain data were compared between treatment groups with Student's test or Wilcoxon's ranks sum test, depending on whether the distribution of quantitative variables could be considered normal or not (QQ plot and Shapiro-Wilk's test), with Chi-square or Fisher's exact test, when appropriate, for qualitative variables.

The between groups comparison of the PIPP over time was conducted with a mixed effect model applied to the rank-transformed data, and the effect was estimated with the relative treatment effect (RTE) and its 95% confidence interval (95% CI) [3].

In the model, we introduced the factors treatment, time, and the treatment by time interaction considering an undetermined matrix of variances and covariance. The statistical significance was set at the 5% level.

The agreement between scales was estimated with Krippendorff's alpha and the 95% confidence interval calculated with the bootstrap method considering 1000 re-samplings, after dichotomizing the pain scores into pain categories, (i.e. absent-mild pain versus moderate to severe pain for each pain scale tested).

All data were analysed using the SAS 9.4 (SAS Institute Inc., Cary, NC, USA) for Windows.

Results

Three-hundred-forty-two preterm infants were admitted to the NICU during the recruitment period (March 2016–May 2017). Of those, 146 were eligible for randomisation and 72 were randomised to groups (Fig. 1).

One patient in the experimental group was excluded from analysis due to the poor quality of video recording, which made it impossible to evaluate the facial expression. There were no statistical differences between the experimental and standard group in relation to patient characteristics at enrolment (except for sex) and in terms of age at the time of procedure and number and type of previous total skin breaking procedures before HP: median 14.0 (min 2.0–max 123.0) versus 12.5 (2.0–96.0), respectively, in the experimental and standard group ($p = 0.7780$) (Table 1).

Inter-rater reliability for pain scales

Because the inter-rater reliability for the pain scales between the two evaluators was high [PIPP 0.989 (0.936–0.998), NIPS

0.995 (0.969–0.999), VAS 0.992 (0.949–0.998), FLACC 0.996 (0.976–0.999)], we used the pain scores attributed to the most expert nurse for group comparison analysis and agreement between scales.

Efficacy of one versus two doses of sucrose for HP

The treatment effect in the experimental group was neither statistically significant nor clinically superior to the standard group ($p = 0.9020$), and there was no interaction with time ($p = 0.2849$), although there was a time effect ($p < 0.0001$) (Fig. 2).

Efficacy of the sucrose dosage

We also analysed the presence of pain intensity by evaluating the severity of pain, defining three categories: (1) No Pain, PIPP < 7.0; (2) mild pain, PIPP between 7.0 and 12.0; and (3) moderate to severe pain, PIPP greater than 12.0 (no responders patients). Only six patients, 8.5% of the study population (four in the experimental and two in the standard group), experienced moderate to severe pain at 30 s (only one patient at 60 s and four at 120 s after HP) (Table 2).

Safety of sucrose administration

No adverse effects were reported, in particular, no episodes of desaturation/bradycardia or “choking” were observed.

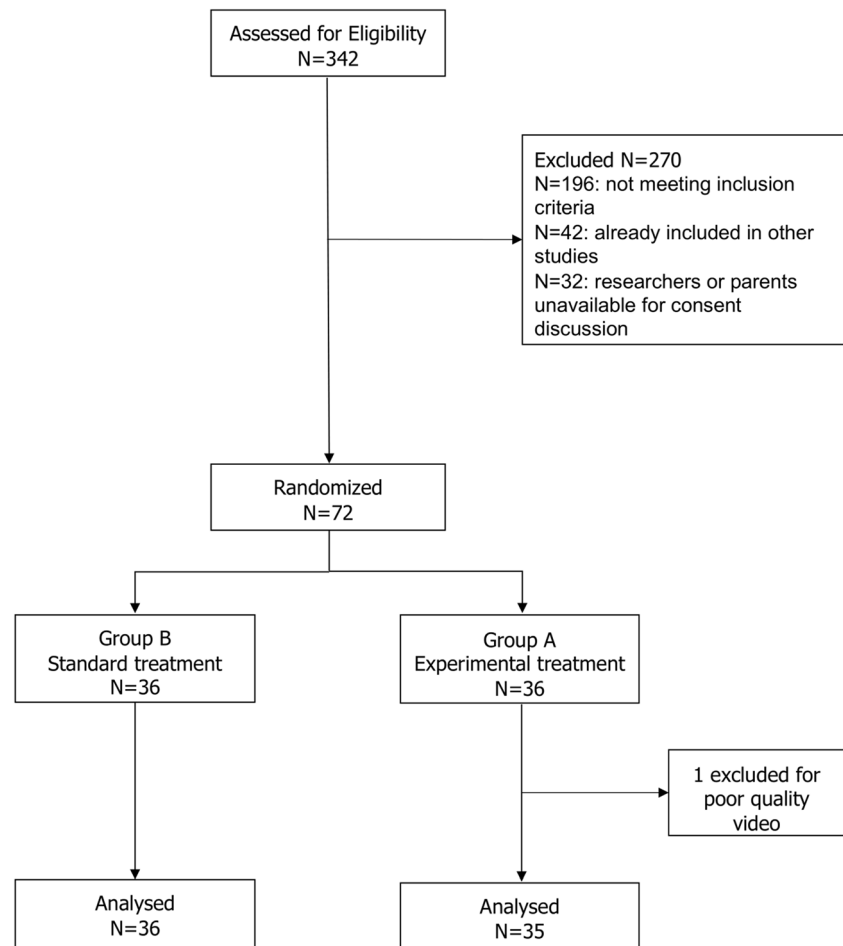
Concordance between pain scales

The analysis was carried out taking into consideration all the times of the procedure. Concordance between PIPP and NIPS was good only at 120 s, while the agreement between PIPP and VAS or FLACC was poor at all times of the procedure (Table 3).

Discussion

This study confirms the efficacy of oral sucrose associated with NNS in controlling HP pain in most of the patients, but our results do not demonstrate any additional benefit when the standard dose is repeated during the HP procedure.

Sucrose is one of the most used non-pharmacological analgesic interventions in newborns. Its efficacy is well-known, and an additive analgesic effect has been reported when sucrose is used in association with NNS and containment (swaddling or FT) [1, 5, 6, 8, 10, 16, 21]. At the time of the study, parents were not allowed to stay during skin breaking procedures, therefore containment were normally performed by a nurse; nowadays, parents are allowed to stay in our NICU 24/7, also during minor procedures (e.g. heel prick, venipuncture,

Fig. 1 Trial flow chart

blood collection), so they can handle the babies during painful stimulus if they wish.

Also, our study demonstrated significant time effects of the treatment, which means the maximum reductions in physiologic and behavioural pain indicators observed at 60 and 120 s from HP, which is evident for both groups. It seemed that the “so-called” analgesic effect of sucrose was primarily mediated by the exposure and that the recruitment of taste sense possibly activates the endogenous opioid system [4, 6, 15, 19], which is in line with the findings of recent studies [2, 22].

Our trial demonstrated that repeating the dose did not change the analgesic effect. Repeated dosing of sucrose was addressed in five studies, but there was no conclusive data [13, 21].

Previous studies have focused on the long term outcomes of cumulative doses of sucrose, and one study has shown an altered motor development evaluation at term in preterm infants exposed to more than ten doses of sucrose per 24 h in the first week of life [11], and another study using a neonatal mouse model has found reduced white and grey matter brain volumes in mouse pups repeatedly exposed to sucrose in the first week of life [24].

However, underestimate and undertreat procedural pain in preterm is no longer acceptable. A review of Valeri et al. found that neonatal pain-related stress was associated with alterations in early and in later developmental outcomes, both for motor and cognitive and emotional outcomes [25].

The mean length of procedures was about 60 s in each arm, considering blood collection and haemostasis. This means that the skin breaking procedure itself had already finished, but the baby was still receiving manipulation while the second dose of sucrose was administered. It is possible that the efficacy of a single sucrose dose is adequate to completely control the pain intensity in short procedures and that a rescue dose could be more effective if the procedure would be longer. However, further studies involving longer procedures could clarify this aspect.

PIPP scale, as for its structure, needs 15 s of observation to evaluate the basal state of the newborn and then 30 s of evaluation after the painful stimulus. In fact, the measurement of PIPP’s parameters for pain evaluation starts right after the procedure and continue during the 30 s after the heel prick. Therefore, we decided to administer the second dose of sucrose or placebo at the end of the pain evaluation to prevent bias.

Table 1 Patient characteristic at enrolment and clinical outcome

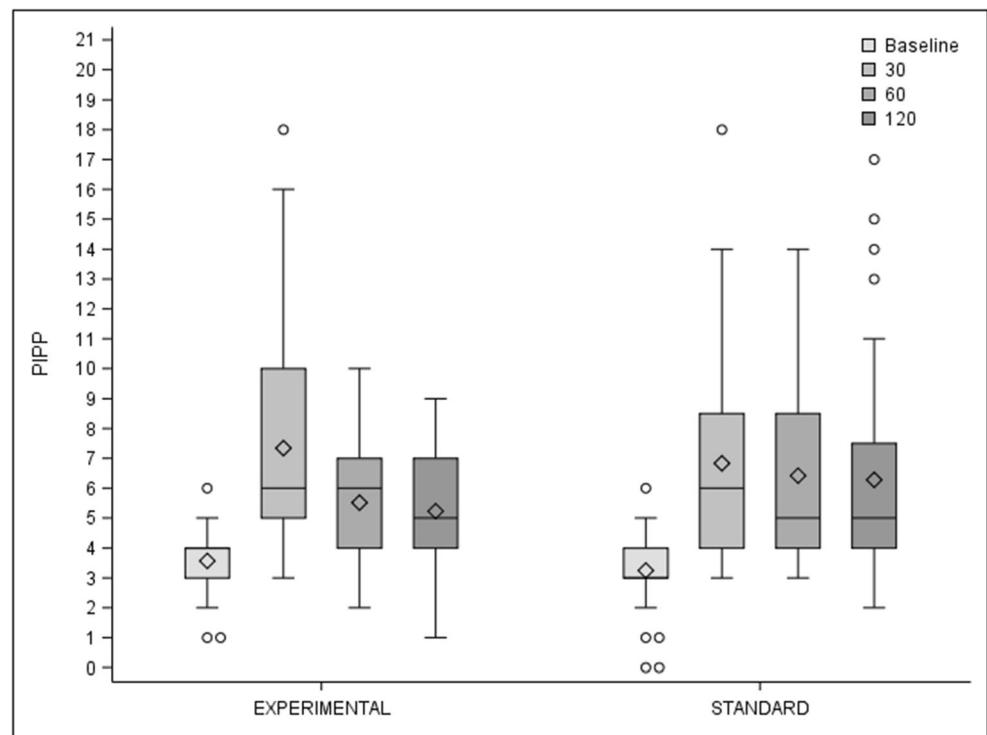
	Experimental group (<i>N</i> = 35)	Standard group (<i>N</i> = 36)	<i>p</i> value
Gestational weeks at birth	28.8	29.4	0.345 ^a
Mean (SD)	(2.7)	(2.8)	
Birth weight (grammes)	1145.0	1232.5	0.967 ^b
Median (min-max)	(660.0–2570.0)	(515.0–2740.0)	
Apgar score 5th minute	8.0 (3.0–10.0)	8.0 (7.0–10.0)	0.408 ^b
Median (min-max)			
CRIB II score	7.0 (1.0–15.0)	5.0 (1.0–14.0)	0.544 ^b
Median (min-max)			
Sex Male <i>n</i> (%)	23 (65.7)	14 (38.9)	0.023^c
Caesarean delivery <i>n</i> (%)	34 (97.1)	30 (83.3)	0.106 ^d
Antenatal steroids <i>n</i> (%)	30 (85.7)	32 (88.9)	0.735 ^d
Gestational weeks at the procedure	31.0 ± 2.7	32.0 ± 2.7	0.117 ^a
Mean ± SD			
Weight at the procedure (g)	5 (14.3)	7 (19.4)	0.562 ^c
< 1000 g <i>n</i> (%)			
Days of life at the procedure	10.0 (1.0–63.0)	15.5 (1.0–60.0)	0.339 ^b
Median (min-max)			
Total heel-prick before the procedure	11.0 (0.0–104.0)	9.5 (0.0–82.0)	0.787 ^b
Median (min-max)			
Total skin-breaking before the procedure	14.0 (2.0–123.0)	12.5 (2.0–96.0)	0.779 ^b
Median (min-max)			
Length of the procedure in seconds	60.8 ± 33.0	59.8 ± 29.9	0.899 ^a
Mean ± SD			
Respiratory support at the procedure <i>n</i> (%)			0.557 ^d
Spontaneous breathing	12 (34.3)	16 (44.4)	
Non-invasive ventilation	19 (54.3)	18 (50.0)	
Mechanical ventilation	4 (11.4)	2 (5.6)	
Outcomes	Experimental group (<i>N</i> = 35)	Standard group (<i>N</i> = 36)	<i>p</i> value
Duration mechanical ventilation (days)	1.0 (0.0–33.0)	0.0 (0.0–48.0)	0.181 ^b
Median (min-max)			
Duration non-invasive ventilation (days)	23.5	13.0	0.292 ^b
Median (min-max)	(0.0–59.0)	(0.0–86.0)	
Need for oxygen at 28 days of life <i>n</i> (%)	11 (36.7)	8 (24.2)	0.283 ^c
Bronchopulmonary dysplasia at 36 weeks <i>n</i> (%)	3 (10.0)	5 (15.6)	0.708 ^d
Patent ductus arteriosus <i>n</i> (%)	11 (36.7)	14 (42.4)	0.640 ^c
Intraventricular haemorrhage grade I–II <i>n</i> (%)	1 (3.3)	4 (12.1)	0.357 ^d
Necrotizing enterocolitis <i>n</i> (%)	2 (6.7)	1 (3.0)	0.601 ^d
Late Onset Sepsis <i>n</i> (%)	6 (20.0)	7 (21.2)	0.905 ^c
Retinopathy of prematurity	4 (13.3%)	5 (16.7%)	1.000 ^d
Hospital stay (days)	45 (10.0–138.00)	36.0 (5.0–143.0)	0.097 ^b

^a Student's *t* test^b Wilcoxon rank-sum test^c Chi-square test^d Fisher's exact test

The sucrose dose interval of 0.3–0.5 mL used in our RCT seemed to be effective in controlling pain in more than 90% of patients, which is in line with most of the international and national consensus and guidelines and

supported by high-quality evidence for the effectiveness [5, 6, 21, 23]. Given the small sample size, our aim was not to compare the two groups (> 1000 g vs < 1000 g) in terms of analgesic efficacy, but only to demonstrate

Fig. 2 Analgesic efficacy of sucrose in the standard and experimental groups



the cumulative effect of the standardized sucrose dosages.

The issue of the effective sucrose dose is still debated, and recently some authors have demonstrated that the minimally effective dose of 24% sucrose required to treat pain associated with HP in preterm infants is 0.1 mL, equal to two drops [22]. Our study was performed prior to the publication of these results and an internal protocol in our NICU fixed the minimum amount of sucrose for HP in 0.3 mL. The findings of Stevens' study will lead to a revision of our protocol reducing the amount of minimum sucrose for HP to 0.1 mL.

In this study, we did not register any adverse events related to sucrose administration. Previous reports confirmed that sucrose administration in the clinical setting was associated with very few adverse events. The literature reports that less than 2% of neonates suffered minor adverse events related to administration of oral solution in general (e.g. choking, brief desaturations), generally rapidly resolved with minimal caregiver intervention [21].

The use of the PIPP scale for pain evaluation in preterm infants has become the gold standard for clinical trials. However, in clinical practice, its use might be difficult: too many parameters must be evaluated within 30 s

Table 2 Frequency of pain intensity by severity at 30 s, 60 s, and 120 s in both groups of treatment

	Experimental group A N. 35	Standard group B N. 36	<i>p</i>
PIPP score at 30 s <i>n</i> (%)			
No pain < 7	22 (62.9)	20 (55.6)	0.438
Mild pain 7–12	9 (25.7)	14 (38.9)	
Moderate–severe Pain > 12	4 (11.4)	2 (5.6)	
PIPP score at 60s <i>n</i> (%)			
No pain < 7	26 (74.3)	23 (63.9)	0.521
Mild pain 7–12	9 (25.7)	12 (33.3)	
Moderate-severe pain > 12	0 (0)	1 (2.8)	
PIPP score at 120 s <i>n</i> (%)			
No pain < 7	25 (71.4)	26 (72.2)	0.094
Mild pain 7–12	10 (28.6)	6 (16.7)	
Moderate-severe pain > 12	0 (0)	4 (11.1)	

Table 3 Concordance between pain scales

PIPP	NIPS	FLACC	VAS
Baseline	Not estimable	− 0.02 (− 1.00–0.74)	− 0.07 (− 0.71–0.5)
30 s	0.32 (− 0.17–0.71)	0.28 (− 0.06–0.57)	0.05 (− 0.22–0.33)
60 s	− 0.01 (− 1.00–0.66)		− 0.11 (− 0.49–0.28)
120 s	0.65 (0.14–1.00)	0.58 (0.16–0.92)	0.48 (0.11–0.78)

by a single operator. For this reason, the standard design of clinical trials involves videotaping and subsequent re-evaluation of the infant's physiological and behavioural reactions. Instead of PIPP in clinical practice, other pain scales might have greater applicability and clinical validity. To evaluate which scales are both easy to apply and precise in the evaluation of pain, we compared the PIPP against two non-validated scales in preterm babies—the FLACC and the indirect VAS—as well as with the NIPS scale, which is validated in premature infants. We obtained a poor concordance between the PIPP and NIPS and between the indirect VAS and FLACC scales.

Strengths and limitations

The methodological strengths of this study are the strategies undertaken to enhance the internal validity of these results. The use of two trained and experienced professionals to perform pain assessment, the randomisation of the sequences to blind the raters to the phase of the procedure, and the use of multiple pain scales reduced potential biases.

We recognise some study limitations. First, we did not stratify our sample according to gestational age. We are aware that gestational age could interfere with pain perception, but this was not the principal aim of our study. Second, the length of the procedure: with the modern HP devices and acquired personnel skills, the procedures lasts no more than 20 s [23]. However, there was an increase in pain score after 2 min, and we could speculate that repeated doses of sucrose would be more effective at controlling prolonged, painful procedures. Finally, we did not conduct a neurological follow-up for the long-term consequences of repeated doses of sucrose in our patients. A higher number of patients would be required to have more conclusive data regarding the comparison of pain scales.

Conclusion

In conclusion, we do not recommend repeating doses of sucrose (24%) during the HP procedure. Further research on the need for a second dose of sucrose in longer skin breaking

procedures is necessary. Other pain scales are not comparable with PIPP, at least for research purposes.

Acknowledgements We thank the patients and their families who participated in this study and the nurses, the residents, the fellows, and the attending physicians of the NICU of University of Padua for having contributed to this study.

Authors' contribution PL conceived the study and wrote the first draught of the manuscript and MEC wrote the final version of the manuscript. TM and AA evaluated the videos in collaboration with VDC; MD elaborated the statistical analysis with the supervision of Prof ACF, who wrote the methodological part of the manuscript. All authors approved the final version of the manuscript.

Compliance with ethical statements

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval The study was approved by the Hospital Ethics Committee (reference number 3714/AO/2016 NEOSUCROSE) and registered in [Clinicaltrials.gov](https://clinicaltrials.gov) (NCT 02859376) before the first patient was enrolled.

Informed consent Informed consent was obtained from all individual participants included in the study.

References

1. Anseloni VC, Ren K, Dubner R, Ennis M (2005) A brainstem substrate for analgesia elicited by intraoral sucrose. *Neuroscience* 133(1):231–243
2. Banga S, Datta V, Rehan HS, Bhakhri BK (2016) Effect of sucrose analgesia, for repeated painful procedures, on short-term neurobehavioral outcome of preterm neonates: a randomized controlled trial. *J Trop Pediatr*. 62(2):101–106
3. Brunner E, Puri ML (2001) Nonparametric methods in factorial designs. *Stat Pap*. 42(1):1–52
4. Carrasco JL, King TS, Chinchilli VM (2009) The concordance correlation coefficient for repeated measures estimated by variance components. *J Biopharm Stat*. 19(1):90–105
5. Cignacco E, Sellam G, Stoffel L, Gerull R, Nelle M, Anand KJ, Engberg S (2012) Oral sucrose and “facilitate tucking” for repeated pain relief in preterms: a randomized controlled trial. *Pediatrics* 129(2):299–308
6. Committee on Fetus and Newborn and Section on Anesthesiology and Pain Medicine (2016) Prevention and management of procedural pain in the neonate: an update. *Pediatrics* 137(2):4271

7. Cruz MD, Fernandes AM, Oliveira CR (2016) Epidemiology of painful procedures performed in neonates: a systematic review of observational studies. *Eur J Pain*. 20(4):489–498
8. Fitzgerald M (2015) What do we really know about newborn infant pain. *Exp Physiol* 100(12):1451–1457
9. Gibbins S, Stevens BJ, Yamada J, Dionne K, Campbell-Yeo M, Lee G, Caddell K, Johnston C, Taddio A (2014) Validation of the premature infant pain profile-revised (PIPP_R). *Early Hum Dev*. 90(4): 189–193
10. Harrison D, Larocque C, Bueno M, Stokes Y, Turner L, Hutton B, Stevens B (2017) Sweet solutions to reduce procedural pain in neonates: a meta-analysis. *Pediatrics* 139(1)
11. Johnston CC, Filion F, Snider L, Majnemer A, Limperopoulos C, Walker CD et al (2002) Routine sucrose analgesia during the first week of life in neonates younger than 31 weeks' postconceptional age. *Pediatrics*. 110(3):523–528
12. Johnston CC, Filion F, Snider L, Limperopoulos C, Majnemer A, Pelousa E et al (2007) How much sucrose is too much sucrose? *Pediatrics*. 119(1):226
13. Kristoffersen L, Malahleha M, Duze Z, Tegnander E, Kapongo N, Støen R, Follestad T, Eik-Nes SH, Bergseng H (2018) Randomized controlled trial showed that neonates received better pain relief from higher dose of sucrose during venipuncture. *Acta Paediatr*. <https://doi.org/10.1111/apa.14567>
14. Lago P, Garetti E, Bellieni CV, Merazzi D, Savant Levet P, Ancora G, Pirelli A, on behalf of the Pain Study Group of the Italian Society of Neonatology (2017) Systematic review of non-pharmacological analgesic interventions for common needle-related procedure in newborn infants and development of evidence-based clinical guidelines. *Acta Paediatr* 106(6):864–870
15. Lawrence J, Alcock D, McGrath P, Kay J, MacMurray SB, Dulberg C (1993) The development of a tool to assess neonatal pain. *Neonatal Netw*. 12(6):59–66
16. Meesters N, Simons S, van Rosmalen J, Reiss I, vanden Aker J, van Dijk M. Waiting 2 minutes after sucrose administration-unnecessary? *Arch Dis Child Fetal Neonatal Ed*. 2017;102(2):F167–F169.
17. Merkel SI, Voepel-Lewis T, Shayevitz JR, Malviya S (1997) The FLACC: a behavioral scale for scoring postoperative pain in young children. *Pediatr Nurs*. 23(3):293–297
18. Ministero della Salute. “Il dolore nel bambino, strumenti pratici di valutazione e terapia.” Available at: www.salute.gov.it. Last access 11/10/2019
19. Ramenghi LA, Evans DJ, Levene MI (1999) “Sucrose analgesia”: absorptive mechanism or taste perception? *Arch Dis Child Fetal Neonatal Ed* 80:F146–F147
20. Stevens B, Johnston C, Petryshen P, Taddio A (1996) Premature infant pain profile: development and initial validation. *Clin J Pain*. 12(1):13–22
21. Stevens B, Yamada J, Ohlsson A, Haliburton S, Shorkey A (2016) Sucrose for analgesia in newborn infants undergoing painful procedures. *Cochrane Database Syst Rev*. 7(7):CD001069
22. Stevens B, Yamada J, Campbell-Yeo M, Gibbins S, Harrison D, Dionne K, Taddio A, McNair C, Willan A, Ballantyne M, Widger K, Sidani S, Estabrooks C, Synnes A, Squires J, Victor C, Riahi S (2018) The minimally effective dose of sucrose for procedural pain relief in neonates: a randomized controlled trial. *BMC Pediatr* 18(1):85–92
23. Taddio A, Yiu A, Smith RW, Katz J, McNair C, Shah V (2009) Variability in clinical practice guidelines for sweetening agents in newborn infants undergoing painful procedures. *Clin J Pain*. 25(2): 153–155
24. Tremblay S, Ranger M, Chau CMY, Ellegood J, Lerch JP, Holsti L, Goldowitz D, Grunau RE (2017) Repeated exposure to sucrose for procedural pain in mouse pups leads to long-term widespread brain alterations. *Pain*. 158(8):1586–1598
25. Valeri BO, Holsti L, Linhares MB (2015) Neonatal pain and developmental outcomes in children born preterm: a systematic review. *Clin J Pain* 31(4):355–362

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